

sonicWORX DSP Plug-In Reference

Complete plug-in list with descriptions — Prosoniq, sonicWORX DSP Function Reference v0.9 (2001)

Gain / Level / Dynamics

DC Offset Correction — Removes DC (direct current) offset from a recording.

Normalize Gain — Scales the loudest peak up to a chosen dB level; optional dithering and U-matic headroom.

Optimize — Boosts level to use all available bits without requantizing (lossless gain).

Shape Volume — Smoothly raises/lowers volume of a selected area by a set amount.

Scale Volume — Scales volume between start/end factors with a variable fade curve; can exceed 0 dB.

Audio Maximizer — Loudness maximizer with auto attack/release; increases punch without distortion.

Fades

Fade-In / Fade-Out — Linear or exponential fade over the selected area.

Exponential Fade-In / Fade-Out — Fade with adjustable curve exponent.

Exponential Fade-curve with Offset — Exponential fade between two defined start/end volumes.

Ease-In Ease-Out Fade-Up / Fade-Down — Cosine-segment ("S-curve") fades.

Volume Ramp-Up / Ramp-Down — Simple linear (ramp) fades.

Channel / Utility

Invert Phase — Mirrors the waveform against the x-axis (180° polarity flip).

Reverse Selection — Reverses playback direction of the selected audio.

Swap Channels — Swaps left and right channels of a stereo file.

Absolute Sample Value — Removes the sign, folding all negative samples positive (full-wave rectify).

Add Channels — Sums inputs A + B with balance control and clipping prevention.

Subtract Channels — Outputs A-B and B-A (cancels mono/center content).

Multiply Channels — Sample-by-sample multiply of A and B (ring-mod-like).

Make Mono — Sums L+R into a mono signal.

Sample Rate Conversion — Converts the sample rate (or just rewrites the header).

Pitch / Time

Pitch Transpose — Changes pitch and length together (resampling).

Natural Pitch Transpose — Pitch transpose with optional timbre/formant preservation.

Phase Vocoder — FFT-based independent time and pitch scaling.

Pitch Scale — Changes pitch without affecting length (optional formant preservation).

Time Scale — Changes length/tempo without affecting pitch.

Pitch Designer — Shifts pitch of components by their tonality; separates instruments in a mix.

Upward / Downward Expansion — Transposes timbral features by harmonic content; length and formants unchanged.

Envelope

Apply Envelope — Transfers the amplitude envelope of one sound onto another.

Invert Envelope — Inverts a sound's envelope (loud↔soft).

Remove Natural Envelope — Strips the amplitude envelope flat (extreme compressor).

EQ / Formant / Spectral

Formant Shaper — Shifts formants only (changes "gender"/vocal character) without pitch change.

Harmonic Contour EQ — Adaptive EQ that alters formant contour and coherency of a voice.

Moving Contour EQ — Same as Harmonic Contour EQ but the effect evolves over time.

Signal (De)Saturation Filter — Adds or removes non-harmonic ("dirt") components.

Wavelet Signal Reduction — Discards least-significant wavelet components for a dirty/dry sound.

Spectral Erosion — Erodes/destroys the spectral shape (good on drums/noise).

Sonic Threshold — Applies a threshold to spectral partials, with randomization.

Vocoders

Vocoder (1) — 18-band analog-style vocoder, dry sound (good for percussion).

Vocoder (2) — Bright, smooth vocoder (good for strings/rich modulations).

Band Reversed Vocoder — Vocoder with reversed-order filter-bank routing (weird/unusual).

Reverb / Delay / Space

APEM De-Reverberation — Removes reverb/room reflections, even from a mono recording (neural/phase-based).

Hall Reverb — Dense, smooth large-room/hall reverb.

Virtual Room — Dense small/medium room reverb (mono-in, stereo-out).

Room Processor — Studio room/hall reverb with reflection-pattern models and warmth/diffusion controls.

Digital Delay — Delay with mix, time, and feedback.

Distortion / Modulation / Special FX

Amplitude Modulation — Modulates amplitude with a sine LFO (with per-channel phase offset).

Center Clipping — Cuts signal below a threshold to the axis, leaving only peaks.

Square Root Distortion — Square-root distortion for "fat" drum loops.

Ring Modulator — Multiplies the signal by a sine carrier for metallic/inharmonic tones.

Transform & Divide — Deconvolves two spectra (filter-impulse-response sounds).

Transform & Multiply — Multiplies two spectra, emphasizing common partials (spectral blend).

(Inverse) Drum Loop Jungelizer — Shifts pitch keeping length/timing for jungle/breakbeat distortion.

Sonic Crowder — Turns spoken voice into a choir of voices.

Sonic Disperse — Fractal feature-distortion (subtle to complex harmonic clusters).

Sonic Electrify — Same fractal engine as Disperse, narrower/more aggressive settings.

Transwave Cycle — Builds a wavetable from the audio and cycles through it (unharmonic enrichment).

Wavelet Alienize — Wavelet-based harmonic distortion with reversed pitch.

Cellular Morph — Morphs between short audio segments for stretched, evolving stereo textures.

Add Stereo Vinyl Noise — Adds old 78-rpm vinyl-style noise.

Morphing / Feature Transfer

Timbral Sound Morphing — Morphs the timbre between two single-instrument sounds.

Structural Sound Morphing — Morphs structural/performance features of single sounds.

Complex Enharmonic Morphing 1 / 2 — Feature-based morphing for complex multi-instrument signals.

Apply Features — Imposes the timbral features of one sound onto another (static "in-between").

Voice Enhancement / Restoration

Vocal Refresh — Multiband exciter for refreshing vocals (also a cheap noise reducer below 100%).

Audio Refresh — Multiband exciter for general audio brightening.

Neural De-click/De-crackle — Neural removal of clicks, pops, and vinyl crackle.

Neural Denoise — Neural broadband noise reduction (best on impulsive material).

Generators

Noise Generator — Generates shaped noise from noise-model files (pitch/metallicity/dispersion).

Sine/Square Generator — Generates a sine or square test tone at a set frequency/amplitude.

Create Noise Model — Produces the noise-model files used by the Noise Generator (referenced only).